

RESEARCH INTEREST

- Model Predictive Control
- Learning for Optimization and Control
- Distributed Optimization
- Multi-Agent Systems
- Connected and Automated Vehicles
- Robotics

EDUCATION

University of Delaware Ph.D. in Mechanical Engineering	Newark, DE USA Aug. 2021–Aug. 2025
Northern Arizona University M.Sc. in Informatics (with Distinction)	Flagstaff, AZ USA Aug. 2019–May 2021
Hanoi University of Science and Technology B.Sc. in Control Engineering and Automation	Hanoi, Vietnam Aug. 2014–Jun. 2019

PROFESSIONAL EXPERIENCE

University of Pennsylvania Postdoctoral Researcher - Department of Electrical and Systems Engineering – Learning to Optimize and Its Applications in Control	Philadelphia, PA USA Aug. 2025–Present
Cornell University Visiting Graduate Scholar - Systems Engineering Field – Learning and Control for Connected and Automated Vehicles in Mixed Traffic – Development of IDS's Scaled Smart City Robotic Testbed (Version 2.0 at Cornell University)	Ithaca, NY USA Aug. 2023–Aug. 2023
Honda Research Institute USA Graduate Student Intern – Social Navigation for Multiple Robots in Crowded Environments	Ann Arbor, MI USA Jun. 2023–Aug. 2023
University of Delaware Graduate Research Assistant - Department of Mechanical Engineering – Learning and Control for Connected and Automated Vehicles in Mixed Traffic	Newark, DE USA Aug. 2021–May 2023
University of Delaware Graduate Teaching Assistant - Department of Mechanical Engineering – Vibration and Control (Lab) - MEEG 312 - Fall 2021 – Dynamics - MEEG 211 - Spring 2022	Newark, DE USA Aug. 2021–May 2022
Northern Arizona University Graduate Research Assistant - School of Informatics, Computing, and Cyber Systems – Learning-Based Model Predictive Control with Gaussian Processes – Adaptive Sampling for Mobile Robotic Sensor Networks	Flagstaff, AZ USA Aug. 2019–May 2021
Vietnam Maritime University Undergraduate Research Intern - School of Mechanical Engineering – Research, Design, and Manufacture of a Floating Crane Testbed in the Laboratory	Haiphong, Vietnam Sep. 2017–Mar. 2019

PUBLICATIONS

JOURNAL ARTICLES

- [1] **V.-A. Le** and A. A. Malikopoulos, “Controller Adaptation via Learning Solutions of Contextual Bayesian Optimization”, *IEEE Robotics and Automation Letters*, vol. 10, no. 8, pp. 8308–8315, 2025.
- [2] **V.-A. Le**, B. Chalaki, F. N. Tzortzoglou, and A. A. Malikopoulos, “Stochastic Time-Optimal Trajectory Planning for Connected and Automated Vehicles in Mixed-Traffic Merging Scenarios”, *IEEE Transactions on Control Systems Technology*, vol. 33, no. 4, pp. 1403–1417, 2025.
- [3] **V.-A. Le** and A. A. Malikopoulos, “Distributed Optimization for Traffic Light Control and Connected Automated Vehicle Coordination in Mixed-Traffic Intersections”, *IEEE Control Systems Letters*, vol. 8, pp. 2721–2726, 2024.
- [4] A. Mokhtarian, P. Scheffe, M. Kloock, S. Schäfer, H. Bang, **V.-A. Le**, S. Sankaramangalam Ulhas, J. Betz, S. Wilson, S. Berman, A. Prorok, and B. Alrifaae, “A Survey on Small-Scale Testbeds for Connected and Automated Vehicles and Robot Swarms”, *IEEE Robotics and Automation Magazine*, 2024.
- [5] A. I. Mahbub, **V.-A. Le**, and A. A. Malikopoulos, “A safety-prioritized receding horizon control framework for platoon formation in a mixed traffic environment”, *Automatica*, vol. 155, p. 111115, 2023.
- [6] **V.-A. Le**, L. Nguyen, and T. X. Nghiem, “Multistep Predictions for Adaptive Sampling in Mobile Robotic Sensor Networks Using Proximal ADMM”, *IEEE Access*, vol. 10, pp. 64 850–64 861, 2022.
- [7] **V.-A. Le**, L. Nguyen, and T. X. Nghiem, “ADMM-Based Adaptive Sampling Strategy for Nonholonomic Mobile Robotic Sensor Networks”, *IEEE Sensors Journal*, vol. 21, no. 13, pp. 15 369–15 378, 2021.
- [8] **V.-A. Le**, X. H. Le, L. Nguyen, and X. M. Phan, “An efficient adaptive hierarchical sliding mode control strategy using neural networks for 3D overhead cranes”, *International Journal of Automation and Computing*, vol. 16, no. 5, pp. 614–627, 2019.
- [9] H. X. Le, T. V. Nguyen, **V.-A. Le**, T. A. Phan, N. H. Nguyen, and M. X. Phan, “Adaptive hierarchical sliding mode control using neural network for uncertain 2D overhead crane”, *International Journal of Dynamics and Control*, vol. 7, pp. 996–1004, 2019.
- [10] H. X. Le, **V.-A. Le**, and L. Nguyen, “Adaptive fuzzy observer based hierarchical sliding mode control for uncertain 2D overhead cranes”, *Cyber-Physical Systems*, vol. 5, no. 3, pp. 191–208, 2019.
- [11] A. T. Le, M. C. Hoang, V. T. Pham, C. N. Luong, D. T. Vu, and **V.-A. Le**, “Adaptive neural network sliding mode control of shipboard container cranes considering actuator backlash”, *Mechanical Systems and Signal Processing*, vol. 112, pp. 233–250, 2018.

CONFERENCE PAPERS

- [1] **V.-A. Le**, R. Tumu, and R. Mangharam, “Toward Single-Step MPPI via Differentiable Predictive Control”, in *2026 IEEE 65th Conference on Decision and Control (CDC)*, (accepted, arXiv preprint arXiv:2604.01539).
- [2] **V.-A. Le** and A. A. Malikopoulos, “Accelerating Time-Optimal Trajectory Planning for Connected and Automated Vehicles with Graph Neural Networks”, in *23rd IFAC World Congress*, (accepted, arXiv preprint arXiv:2511.20383).
- [3] **V.-A. Le**, M. Xie, and R. Mangharam, “A Hybrid Learning-to-Optimize Framework for Mixed-Integer Quadratic Programming”, in *Proceedings of the 8th Annual Learning for Dynamics and Control Conference*, ser. Proceedings of Machine Learning Research, vol. 331, PMLR, 2026, pp. 1055–1069.
- [4] A. Amine, K. Puri, **V.-A. Le**, and R. Mangharam, “Nonplanar Model Predictive Control for Autonomous Vehicles with Recursive Sparse Gaussian Process Dynamics”, in *2026 IEEE Intelligent Vehicles Symposium (IV)*, IEEE, 2026, pp. 2088–2093.
- [5] **V.-A. Le**, P. Kounatidis, and A. A. Malikopoulos, “Combining Graph Attention Networks and Distributed Optimization for Multi-Robot Mixed-Integer Convex Programming”, in *2025 IEEE 64th Conference on Decision and Control (CDC)*, IEEE, 2025, pp. 4440–4445.
- [6] **V.-A. Le**, B. Chalaki, V. Tadiparthi, H. N. Mahjoub, J. D’Sa, and E. Moradi-Pari, “Social Navigation in Crowded Environments with Model Predictive Control and Deep Learning-Based Human Trajectory Prediction”, in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 4793–4799.
- [7] **V.-A. Le**, V. Tadiparthi, B. Chalaki, H. N. Mahjoub, J. D’sa, E. Moradi-Pari, and A. A. Malikopoulos, “Multi-Robot Cooperative Navigation in Crowds: A Game-Theoretic Learning-Based Model Predictive Control Approach”, in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2024, pp. 4834–4840.

- [8] **V.-A. Le**, H. M. Wang, G. Orosz, and A. A. Malikopoulos, “Coordination for Connected Automated Vehicles at Merging Roadways in Mixed Traffic Environment”, in *2023 62nd IEEE Conference on Decision and Control (CDC)*, 2023, pp. 4150–4155.
- [9] N. Venkatesh, **V.-A. Le**, A. Dave, and A. A. Malikopoulos, “Connected and Automated Vehicles in Mixed-Traffic: Learning Human Driver Behavior for Effective On-Ramp Merging”, in *2023 62nd IEEE Conference on Decision and Control (CDC)*, 2023, pp. 92–97.
- [10] **V.-A. Le** and A. A. Malikopoulos, “Optimal Weight Adaptation of Model Predictive Control for Connected and Automated Vehicles in Mixed Traffic with Bayesian Optimization”, in *2023 American Control Conference (ACC)*, 2023, pp. 1183–1188.
- [11] **V.-A. Le** and A. A. Malikopoulos, “A Cooperative Optimal Control Framework for Connected and Automated Vehicles in Mixed Traffic Using Social Value Orientation”, in *2022 61st IEEE Conference on Decision and Control (CDC)*, 2022, pp. 6272–6277.
- [12] A. M. I. Mahbub, **V.-A. Le**, and A. A. Malikopoulos, “Safety-Aware and Data-Driven Predictive Control for Connected Automated Vehicles at a Mixed Traffic Signalized Intersection”, in *10th IFAC International Symposium on Advances in Automotive Control*, IFAC, 2022, pp. 51–56.
- [13] **V.-A. Le** and T. X. Nghiem, “Distributed Experiment Design and Control for Multi-agent Systems with Gaussian Processes”, in *2021 60th IEEE Conference on Decision and Control (CDC)*, 2021, pp. 2226–2231.
- [14] **V.-A. Le** and T. X. Nghiem, “A Receding Horizon Approach for Simultaneous Active Learning and Control using Gaussian Processes”, in *2021 IEEE Conference on Control Technology and Applications (CCTA)*, IEEE, 2021, pp. 453–458.
- [15] **V.-A. Le**, L. Nguyen, and T. X. Nghiem, “An Efficient Adaptive Sampling Approach for Mobile Robotic Sensor Networks using Proximal ADMM”, in *2021 American Control Conference (ACC)*, IEEE, 2021, pp. 1101–1106.
- [16] **V.-A. Le** and T. X. Nghiem, “Gaussian Process Based Distributed Model Predictive Control for Multi-agent Systems using Sequential Convex Programming and ADMM”, in *2020 IEEE Conference on Control Technology and Applications (CCTA)*, IEEE, 2020, pp. 31–36.
- [17] T. X. Nghiem, T.-D. Nguyen, and **V.-A. Le**, “Fast Gaussian Process based Model Predictive Control with Uncertainty Propagation”, in *2019 57th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, IEEE, 2019, pp. 1052–1059.
- [18] **V.-A. Le**, X. H. Le, D. T. Vu, V. T. Pham, A. T. Le, and M. C. Hoang, “Designing an adaptive controller for 3D overhead cranes using hierarchical sliding mode and neural network”, in *2018 International Conference on System Science and Engineering (ICSSE)*, IEEE, 2018, pp. 1–6.

FELLOWSHIPS AND AWARDS

- Jun. 2026: IEEE ITSS Best Dissertation Award - 2nd Place by the IEEE Intelligent Transportation Systems Society
- Feb. 2026: Penn AI Fellowship by the University of Pennsylvania
- Jul. 2025: Best Paper Finalist by the ASME Automotive and Transportation Systems Technical Committee (ATS-TC) for the 2025 American Control Conference
- Aug. 2023: Student Travel Award by IEEE Control Systems Society for the 2023 IEEE Conference on Decision and Control
- Oct. 2022: IEEEExtreme Programming Competition - Global Rank: 949/6373, US Rank: 12/99
- Apr. 2022: Student Travel Awards by the University of Delaware’s Graduate College and IEEE Control Systems Society for the 2022 American Control Conference
- Sep. 2021: Student Travel Award by IEEE Control Systems Society for the 2021 IEEE Conference on Decision and Control
- Jun. 2021: Student Travel Award by IEEE Control Systems Society for the 2021 IEEE Conference on Control Technology and Applications
- Aug. 2020: Student Travel Award by IEEE Control Systems Society for the 2020 IEEE Conference on Control Technology and Applications
- Aug. 2019: Northern Arizona University’s Presidential Fellowship

- Aug. 2018: Odon Vallet’s Scholarship (established by Prof. Odon Vallet from Sorbonne University) for undergraduate students
- Jun. 2018: Conference Travel Award by Vietnam’s National Foundation for Science and Technology Development (NAFOSTED) for the 2018 IEEE International Conference on System Science and Engineering
- Apr. 2015: Gold Medal in the 2015 Vietnam’s National Mathematical Olympiad for undergraduate students

SEMINARS AND TALKS

1. October 2026: The IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Tutorial on “Learning-to-Optimize for Control in Robotics.” (Organizer and Speaker)
2. May 2026: The 3rd Northeast Systems and Control Workshop, “Learning-to-Optimize and Its Applications in Model Predictive Control.”
3. May 2025: RGSO-CARS Seminar at the University of Delaware, “Towards Real-time Cooperative Trajectory Planning for Connected and Automated Vehicles in Mixed Traffic.”

ACADEMIC SERVICES

- Membership

– Member, Institute of Electrical and Electronics Engineers (IEEE)	2020–Present
– Member, IEEE Control System Society	2020–Present
– Member, IEEE Intelligent Transportation Systems Society	2022–Present
– Member, IEEE Robotics and Automation Society	2022–Present
– Member, IEEE-CSS Technical Committee on Smart Cities	2022–Present
– Member, IEEE-CSS Technical Committee on Automotive Controls	2024–Present

- Reviewer

- Journals: Automatica, IEEE Transactions on Automatic Control; IEEE Transactions on Intelligent Transportation Systems; Transportation Research Part C: Emerging Technologies; IEEE/ASME Transactions on Mechatronics; IEEE Transactions on Robotics; IEEE Control Systems Letters; Journal of the Franklin Institute; IEEE Transactions on Control Systems Technology; IEEE Transactions on Vehicular Technology; IEEE Transactions on Intelligent Vehicles, IEEE Open Journal of Control Systems, IEEE Robotics and Automation Letters
- Conferences: IEEE Conference on Control Theory and Applications; IEEE Conference on Decision and Control; American Control Conference; European Control Conference; IEEE International Conference on Intelligent Transportation Systems; IEEE International Conference on Robotics and Automation; IEEE/RSJ International Conference on Intelligent Robots and Systems; IEEE Intelligent Vehicles Symposium; Modeling, Estimation and Control Conference, IFAC World Congress